DO NOW — RECALL

Lesson 110 · Round Robin and Multilevel Feedback Queu

Name: _____ Class: _____ Date: _____

1. Which scheduler preempts a running task when a shorter one arrives?

2. Why can an array provide direct indexed access?

3. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

4. For the binary value 1010, what is its decimal value?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Shortest remaining time (SRT).

2. Model answer

Its element locations can be calculated from base address and index.

3. Model answer

Sum 0 and carry-out 1.

4. Model answer

10.

By the end, you can…

• Ready processes receive a time quantum in cyclic order.

• Expired slices trigger a switch; very small slices create overhead.

• Queues with differing priorities/quanta adapt to process behaviour.

Vocabulary — know these terms

time quantum — Maximum continuous CPU interval per RR turn.

round robin — Preemptive cyclic scheduling.

context switch — Saving/restoring CPU state when changing process.

MLFQ — Multilevel feedback queue scheduling.

NEW LEARNING

Read, annotate and explain

Round robin

Ready processes receive a time quantum in cyclic order.

Context switching

Expired slices trigger a switch; very small slices create overhead.

MLFQ

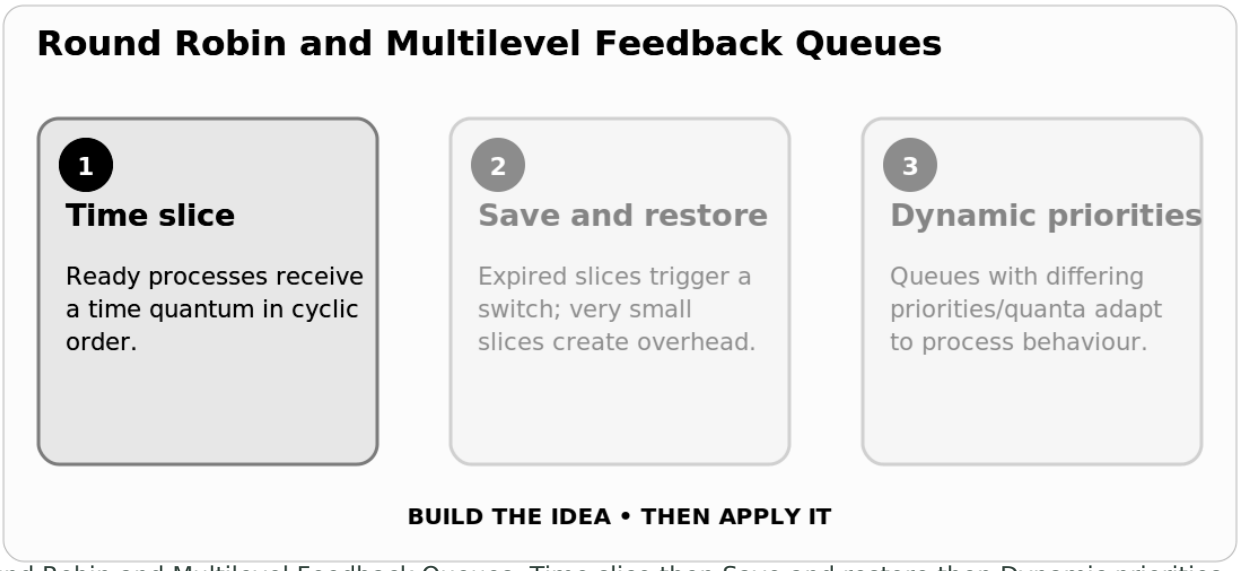
Queues with differing priorities/quanta adapt to process behaviour.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

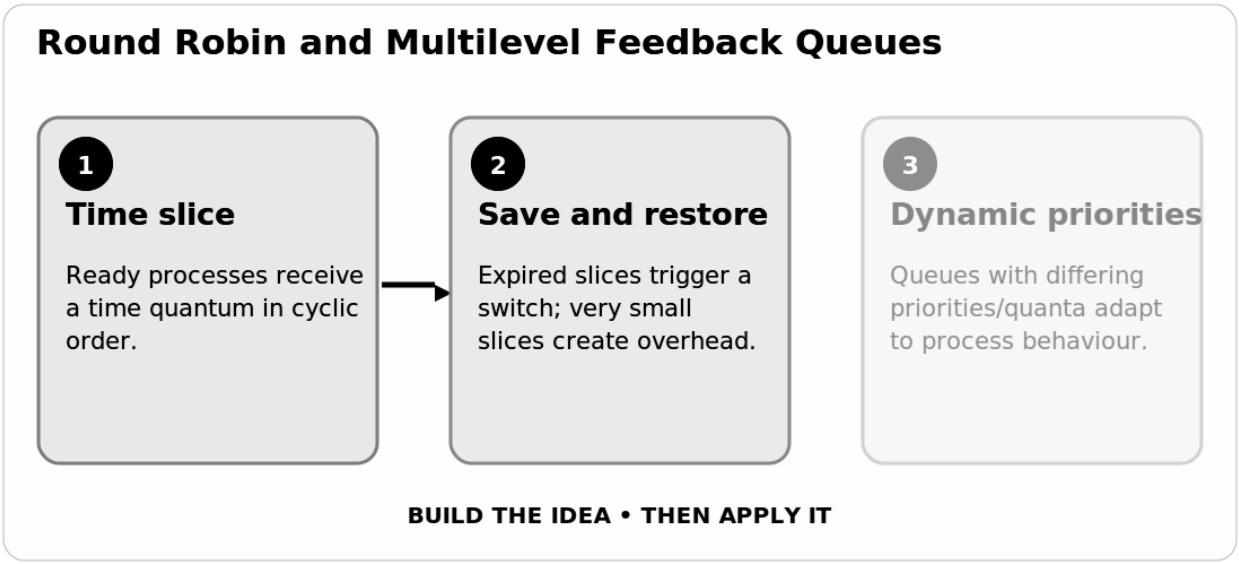
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Round Robin and Multilevel Feedback Queues: Time slice then Save and restore then Dynamic priorities

New Learning

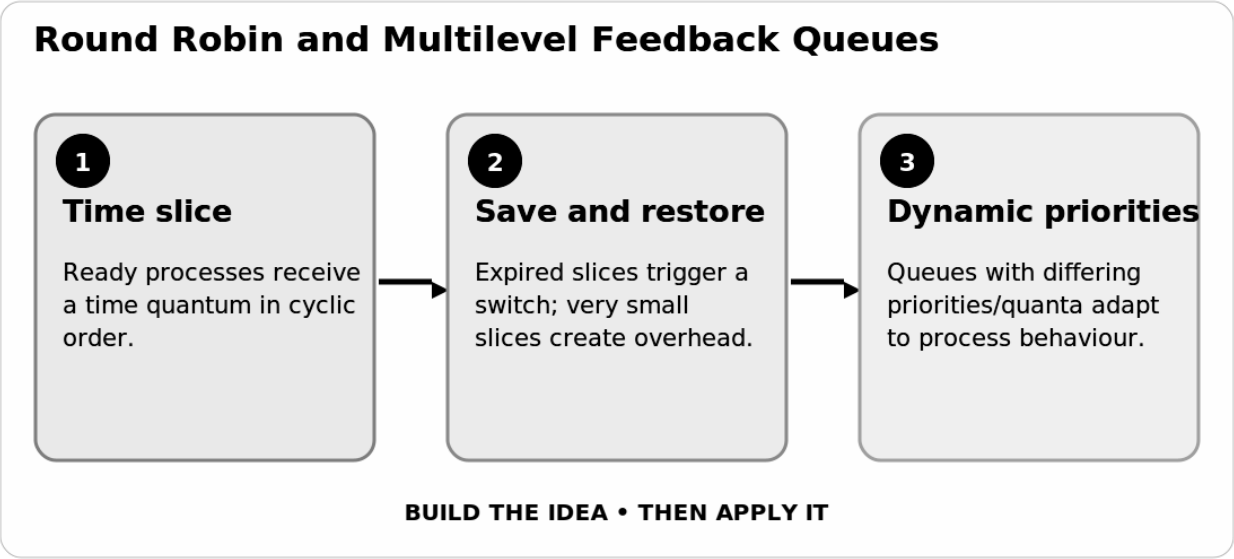


Round Robin and Multilevel Feedback Queues: Time slice then Save and restore then Dynamic priorities

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Round Robin and Multilevel Feedback Queues: Time slice then Save and restore then Dynamic priorities

I DO — WORKED EXAMPLE

Follow the worked method

RR with quantum 2: P1 burst 5, P2 burst 3; both arrive at t=0. Trace completion.

Model answer

P1 runs 0–2, P2 2–4, P1 4–6, P2 6–7 (finishes), P1 7–8 (finishes). Completion times: P2 7, P1 8.

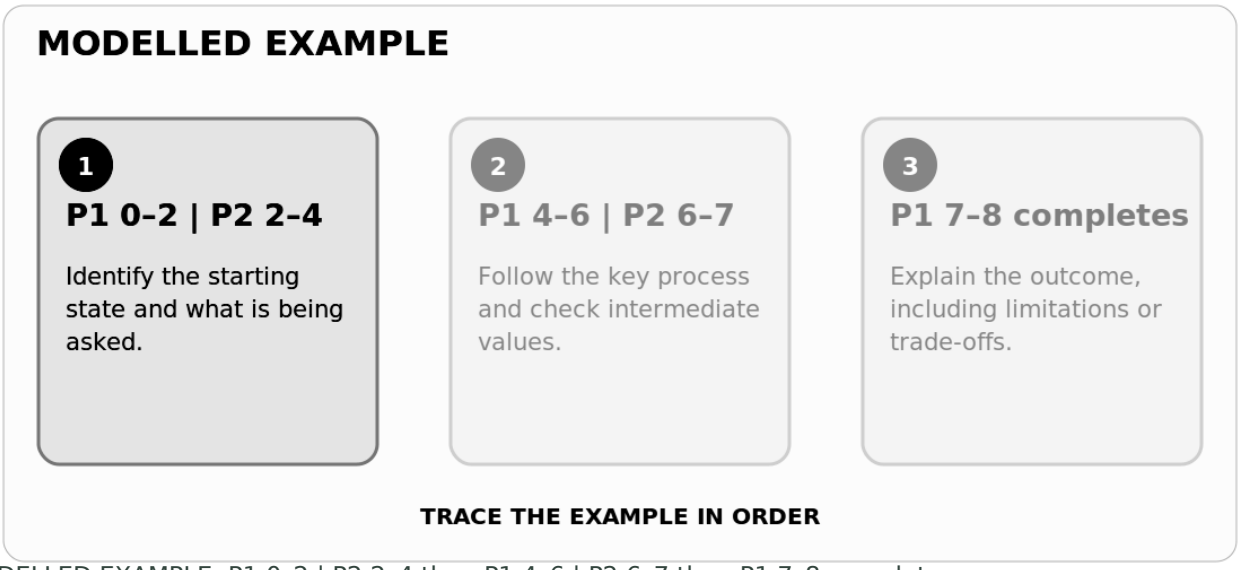
Check: link each claim to the scenario and use precise OCR terminology.

I Do · original lesson diagrams

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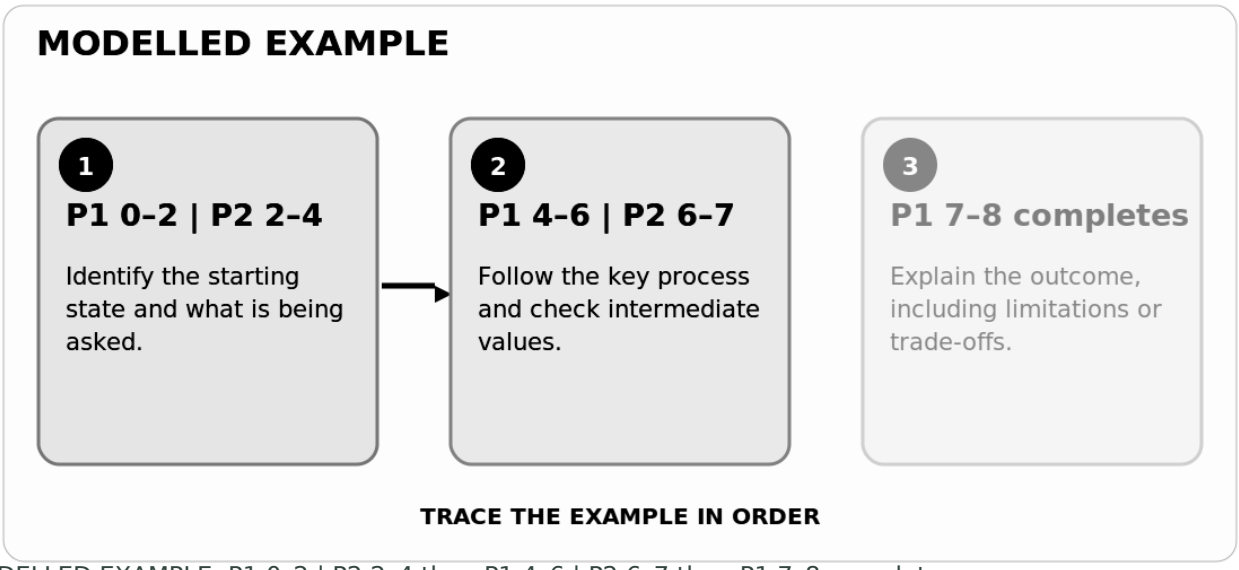
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MODELLED EXAMPLE: P1 0–2 | P2 2–4 then P1 4–6 | P2 6–7 then P1 7–8 completes

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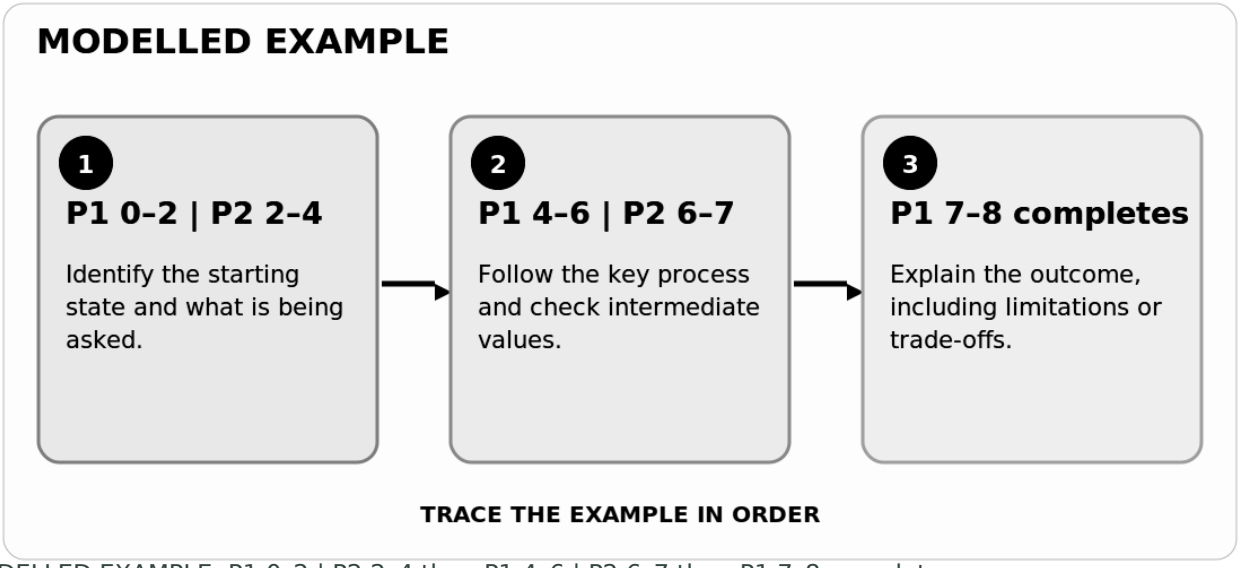
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WE DO — GUIDED PRACTICE

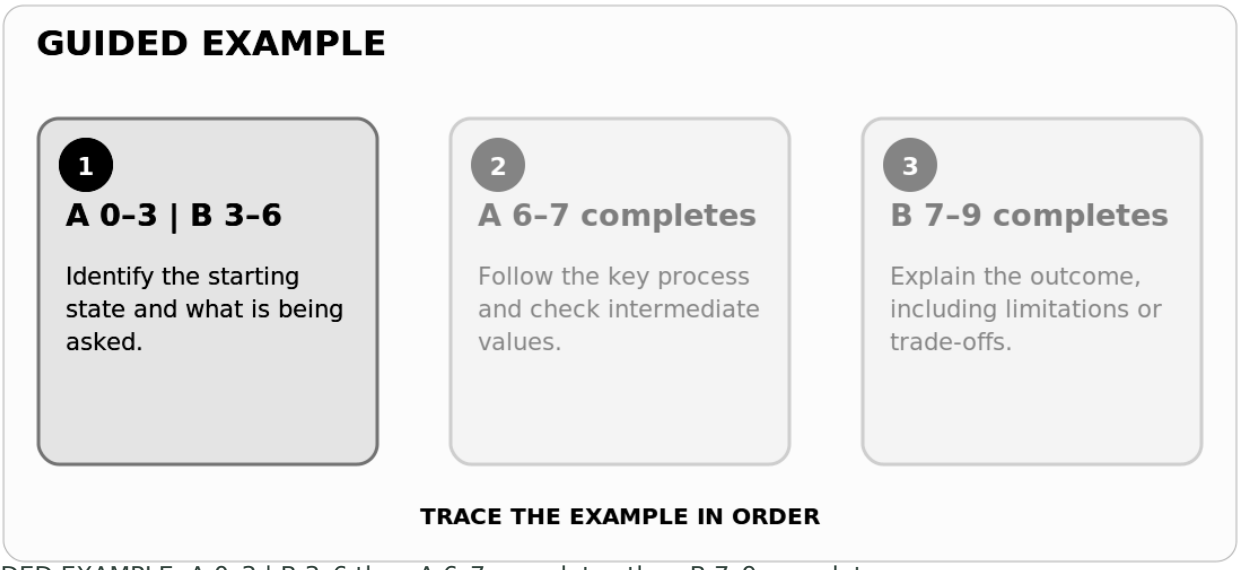
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

RR quantum 3: A burst 4, B burst 5; both at time 0. Trace the run order.

Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: A 0–3 | B 3–6 then A 6–7 completes then B 7–9 completes

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe the round-robin algorithm, including the purpose of the time quantum. [6]

Question 2

RR quantum 2: A burst 3, B burst 4, both arrive 0. Produce a Gantt timeline. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain how MLFQ responds to CPU-bound and interactive tasks and evaluate a very short quantum. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Reducing a time quantum to an extremely small value always improves performance.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

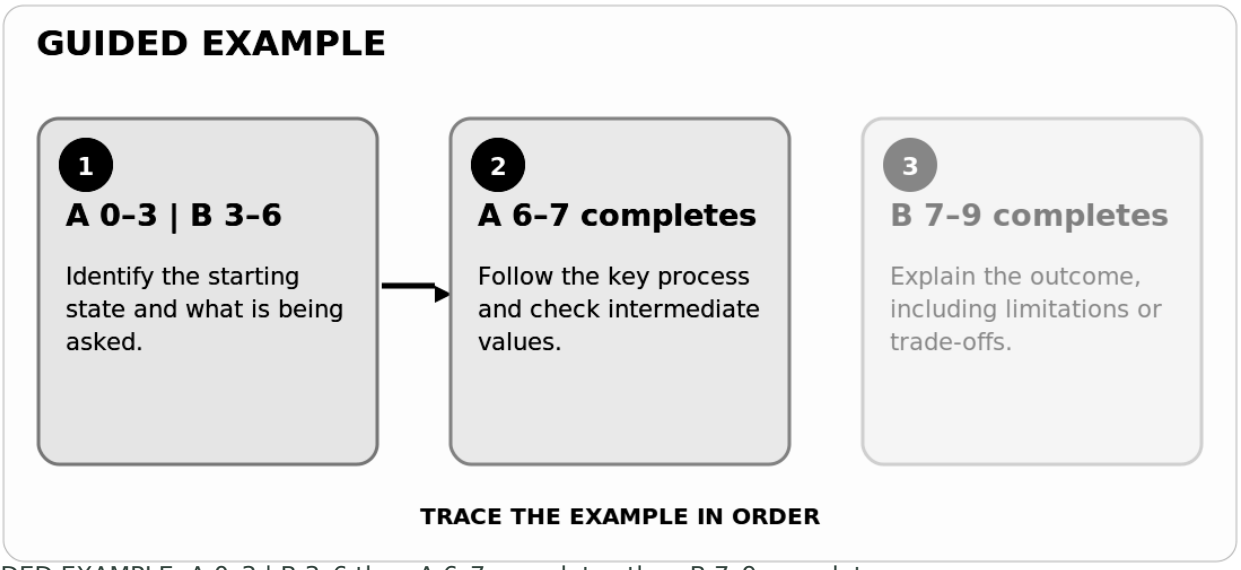
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

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RR quantum 3: A burst 4, B burst 5; both at time 0. Trace the run order.

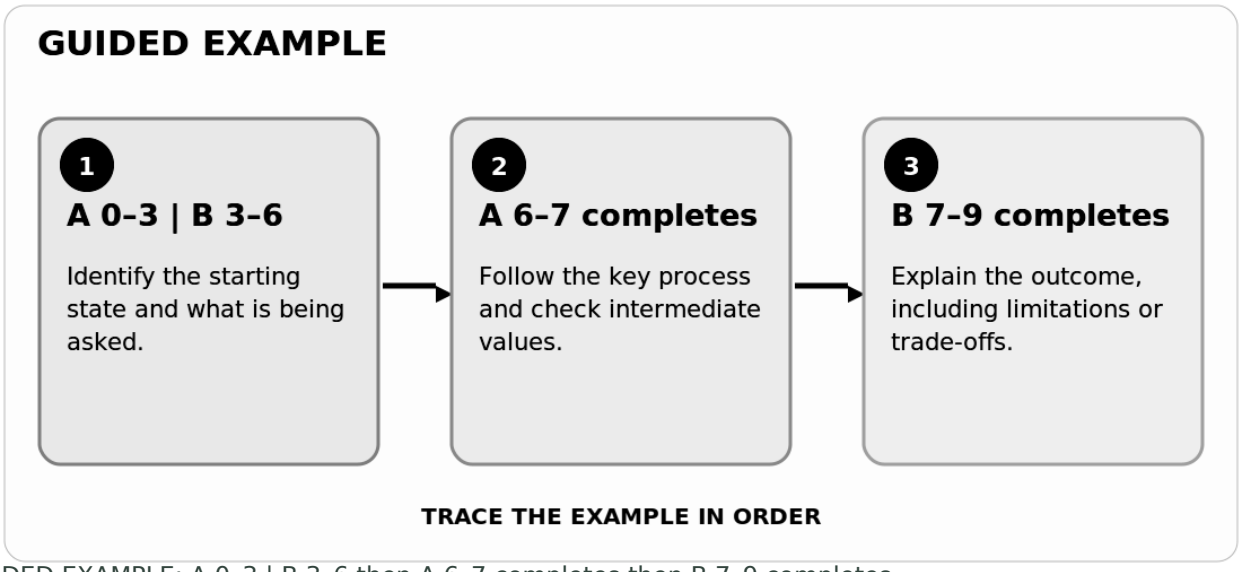
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GUIDED EXAMPLE: A 0–3 | B 3–6 then A 6–7 completes then B 7–9 completes

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Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: A 0–3 | B 3–6 then A 6–7 completes then B 7–9 completes

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

A 0–3 (1 left), B 3–6 (2 left), A 6–7 finishes, B 7–9 finishes. Turnaround A 7 and B 9.

IMPROVE: Explain one correction to your own reasoning.